

Neomycin 528.5 mg/g Water Soluble Powder

Neomycin 528.5 mg/g Water Soluble Powder appears to be a light yellow color, crystalline powder.

Composition:

Each g contains

Neomycin Sulfate

755 mg

Pharmacodynamics:

Aminoglycosides enter susceptible bacteria by oxygen-dependent active transport (making anaerobes impervious to them) and by passive diffusion. Once the antibiotic has gained access, it binds irreversibly to a receptor protein on the 30S ribosomal subunit and blocks the formation of a complex that includes mRNA, formylmethionine, and tRNA. As a result, the tRNA is translated incorrectly, producing a nonfunctional protein. Aminoglycosides also disrupt protein synthesis by disruption of polysomes and may prevent the initiation of DNA replication.

Pharmacokinetics:

In general, aminoglycosides and apramycin are very poorly absorbed from oral administration in adult animals, including cattle, chickens, and pigs. Aminoglycosides are distributed primarily into the extracellular space and over time accumulate in tissues. The amount of antibiotic in most tissues appears to be dependent on the total dose administered over time rather than the size of each individual dose. Aminoglycosides do not distribute well across membrane barriers and, therefore, are not found at high concentrations in brain tissue, cerebrospinal fluid, ocular fluid, or respiratory secretions.

Indication:

For the treatment and control of colibacillosis (bacterial enteritis) caused by *Escherichia coli* susceptible to neomycin sulfate in cattle (excluding veal calves), swine, sheep and goats. For the control of mortality associated with *Escherichia coli* organisms susceptible to neomycin sulfate in growing turkeys.

Recommended Dosage:

Administer to cattle (excluding veal calves), swine, sheep and goats at a dose of 10 mg neomycin sulfate per 0.45 kg of body weight per day in divided doses for a maximum of 14 days. Administer to turkeys at a dose of 10 mg neomycin sulfate per 0.45 kg of body weight per day for 5 days.

HERD/FLOCK TREATMENT:

100 g Neomycin 528.5 mg/g Water Soluble Powder will treat 3397.5 kg of body weight. Estimate the total number of kg of body weight of the animals to be treated and administer 100 g for each 3397.5 kg. The product should be added to the amount of drinking water estimated to be consumed in 12-24 hours. Provide medicated water as the sole source of water each day until consumed, followed by non-medicated water as required. Fresh medicated water should be prepared each day.

Mode of Administration:

Oral Administration.

Contraindications:

The medical contraindications have been selected on the basis of their potential clinical significance. Except under special circumstances, this medication should not be used when the following medical problems exist:

- Dehydration, hypovolemic
- Hypersensitivity to aminoglycosides
- Renal dysfunction

Warnings and Precautions:

To administer the stated dosage, the concentration of neomycin required in medicated water must be adjusted to compensate for variation in age and weight of animal, the nature and severity of disease signs, and environmental temperature and humidity, each of which affects water consumption.

If symptoms persist after using this preparation for 2 or 3 days, consult a veterinarian. If symptoms such as fever, depression, or going off feed develop, oral neomycin is not indicated as the sole treatment since systemic levels of neomycin are not obtained due to low absorption from the gastrointestinal tract.

Treatment should continue 24 to 48 hours beyond remission of disease symptoms, but not to exceed a total of 14 consecutive days for cattle, swine, sheep and goats and 5 days for turkeys.

Interactions with Other Medicaments:

The following drug interactions have been selected on the basis of their potential clinical significance. Combinations containing any of the following medications, depending on the amount present, may also interact with this medication.

- Aminoglycosides, two or more concurrently
- Calcium Channel blockers
- Calcium
- Iron
- Nephrotoxic medications
- Ototoxic medications

Statement on Usage During Pregnancy and Lactation:

Because of poor lipid solubility, aminoglycosides have relatively poor penetration from plasma into milk.

Adverse Effects / Undesirable Effects:

The following side/adverse effects have been selected on the basis of their potential clinical significance: Nephrotoxicity, ototoxicity, auditory; ototoxicity, vestibular.

Overdose and Treatment:

For neuromuscular blockade

- Administration of edrophonium, 0.5 mg/kg, will reverse neuromuscular blocking effects. Administration of calcium chloride at 10 to 20 mg /kg, calcium gluconate at 30 to 60 mg/kg, or neostigmine at a dose of 100 to 200 mcg per kg of body weight can also reverse muscle response depression and associated dyspnea.

For renal toxicity

- Aminoglycoside administration should be immediately discontinued.
- Polyionic electrolyte fluid therapy should be initiated to stimulate diuresis.

Withdrawal Period (s):

DISCONTINUE TREATMENT PRIOR TO SLAUGHTER BY AT LEAST THE NUMBER OF DAYS LISTED BELOW FOR APPROPRIATE SPECIES:

Cattle.....	1 day
Swine and Goats.....	3 days
Sheep.....	2 days
Turkeys.....	0 days

A withdrawal period has not been established for this product in pre-ruminating calves. Do not use in calves to be processed for veal.

A milk discard period has not been established for this product in lactating dairy cattle. Do not use in female dairy cattle 20 months of age or older.

Storage Conditions:

Store below 30°C. Protect from heat and sunlight exposure.

Keep medicine out of reach of children. (Jauhkan ubat dari kanak-kanak).

Any unused veterinary medicinal product or waste materials derived from such veterinary medicinal products should be disposed of in accordance with local requirements.

Shelf-life (packaged for sale): 3 years

Shelf-life (after first opening of container): 7 days.

Shelf-life (after reconstitution/dilution): 24 hours.

Packaging Available:

100 g & 1 kg

Date of revision:

04-11-2015

Product Registration Holder And Manufacturer By:

NAM PHARMA SDN BHD

5, Lebuhr Perusahaan Klebang 11,

Taman Perindustrian Antarabangsa IGB,

31200 Chemor, Perak, Malaysia.

Tel: +605-2931838, 1839 Fax: +605-2931833